

CATS⁺

Competishun Advance

Training Sessions

(Lecture - III)

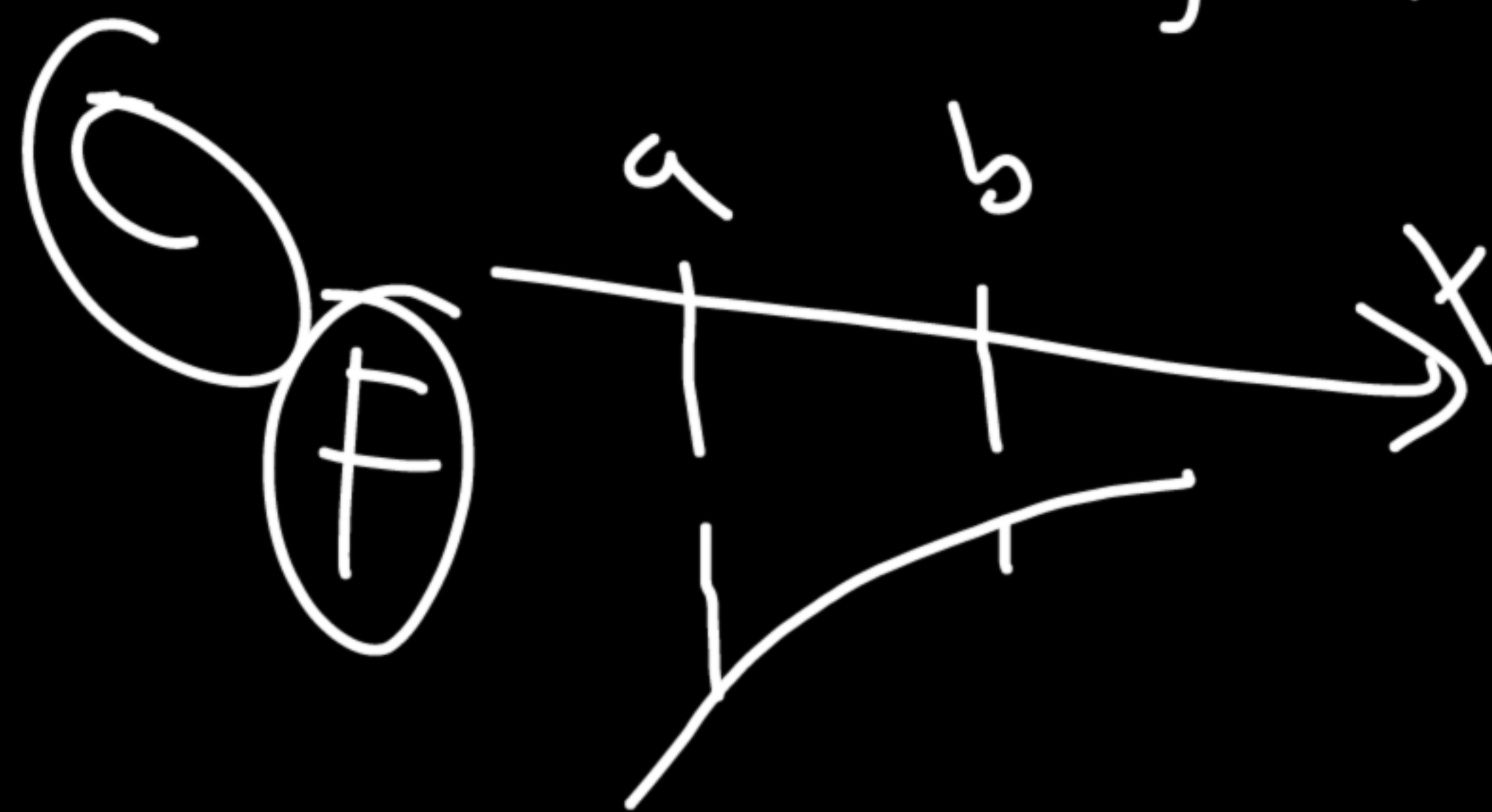
Ques Consider a real valued continuous function $f(x)$ defined on the interval $[a, b]$. Which of the following statements does not hold(s) good?

- ✓ (A) If $f(x) \geq 0$ on $[a, b]$ then $\int_a^b f(x) dx \leq \int_a^b f^2(x) dx$.
- ✓ (B) If $f(x)$ is increasing on $[a, b]$, then $f^2(x)$ is increasing on $[a, b]$.
- ✓ (C) If $f(x)$ is increasing on $[a, b]$, then $f(x) \geq 0$ on (a, b) .
- ✓ (D) If $f(x)$ attains a minimum at $x = c$ where $a < c < b$, then $f'(c) = 0$.

(A) $f(x) \geq 0$, $\int_a^b f(x) dx \leq \int_a^b f^2(x) dx$ (F)

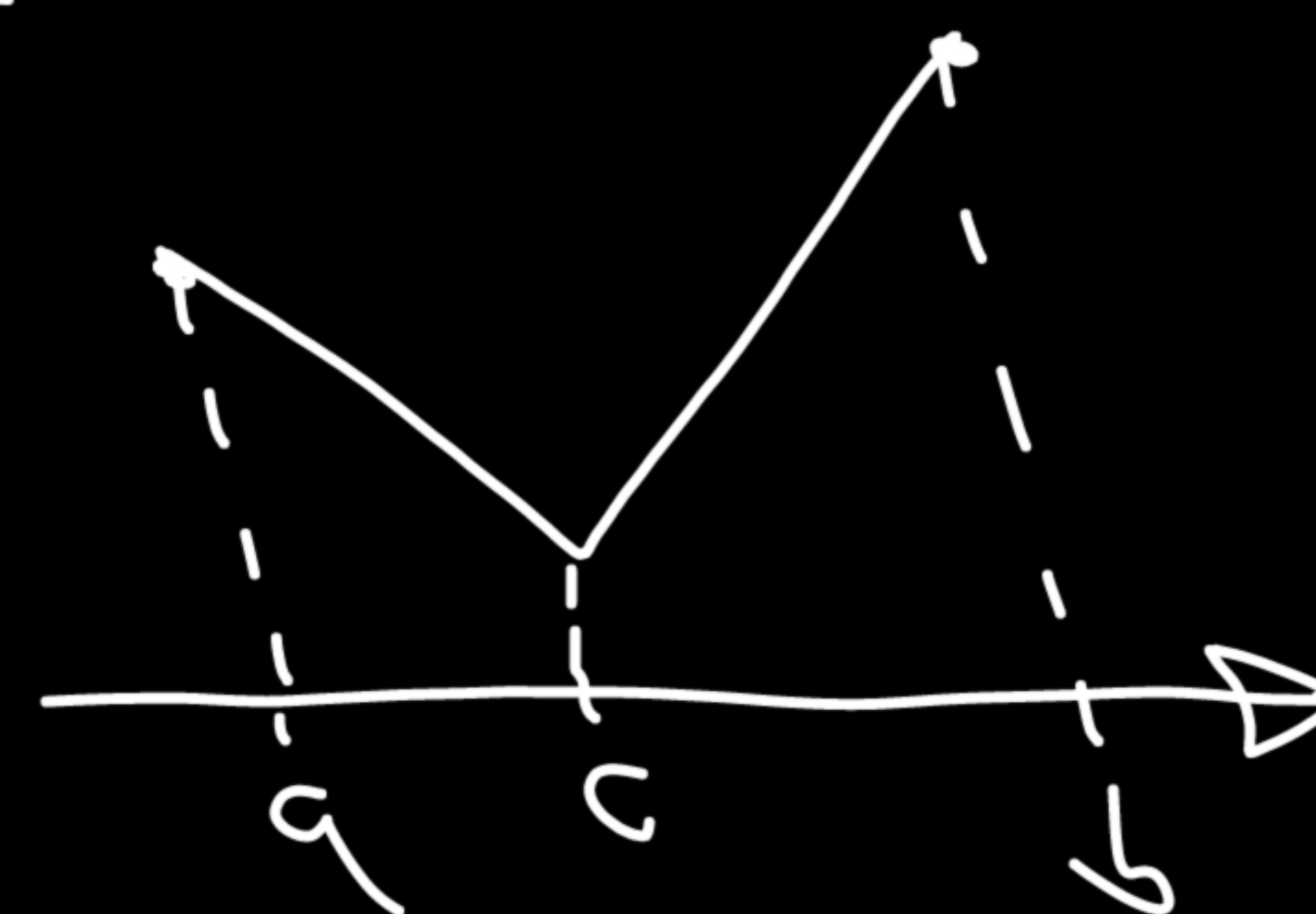
(B) (F)

$$\begin{aligned} f(x) &= x \\ f^2(x) &= x^2 \end{aligned}$$



fractional part f

(D) (F)



Ques

Let z_1, z_2, z_3 are three complex numbers such that $|z_1| = |z_2| = |z_3| = 1$ and

$$\frac{z_1^2}{z_2 z_3} + \frac{z_2^2}{z_1 z_3} + \frac{z_3^2}{z_1 z_2} + 1 = 0 \text{ then integral values that } |z_1 + z_2 + z_3| \text{ can assume is}$$

(A) 1

(B) 2

(C) 3

(D) 4

$$\frac{z_1^2}{z_2 z_3} + \frac{z_2^2}{z_1 z_3} + \frac{z_3^2}{z_1 z_2} + 1 = 0.$$

$$\Rightarrow z_1^3 + z_2^3 + z_3^3 + z_1 z_2 z_3 = 0.$$

$$\Rightarrow z_1^3 - 3z_1 = -4z_1$$

$$\Rightarrow (\sum z_1)(\sum z_1^2 - \sum z_1 z_2) = -4z_1$$

$$\Rightarrow (\sum z_1)(\sum z_1^2 - 3 \sum z_1 z_2) = -4z_1$$

$$(\sum z_1)^3 - 3(\sum z_1)(\sum z_1 z_2) = -4z_1$$

$\Rightarrow$

$$|z_1 + z_2 + z_3|$$

$$|z_1| = |z_2| = |z_3| = 1$$

$$z_1 = \frac{1}{\bar{z}_1}$$

$$z_2 = \frac{1}{\bar{z}_2}$$

$$z_3 = \frac{1}{\bar{z}_3}$$

$$(\sum z_i)^3 = 3(\sum z_i)(\sum z_1 z_2) - 4 \prod z_i$$

$$\hookrightarrow \underline{z_1 z_2 z_3} \left(\frac{1}{z_3} + \frac{1}{z_1} + \frac{1}{z_2} \right) \rightarrow \frac{\overline{z_3} + \overline{z_1} + \overline{z_2}}{\underline{z_1 + z_2 + z_3}}$$

$$= (\prod z_i) \left(3(\sum z_i)(\overline{\sum z_i}) - 4 \right)$$

$$(\sum z_i)^3 = (\prod z_i) (3 |\sum z_i|^2 - 4)$$

$$|| \Rightarrow |z_1 + z_2 + z_3|^3 = \cancel{|z_1 z_2 z_3|} \cdot |3 |z_1 + z_2 + z_3|^2 - 4|$$

$$\Rightarrow t^3 = |3t^2 - 4|$$

$$\hookrightarrow t = \underset{\checkmark}{1}, \underset{\checkmark}{2}, \underset{\times}{3}, \underset{\times}{4}$$

Let
 $|z_1 + z_2 + z_3| = t$

One

In $\triangle ABC$, angle A is 120° , $BC + CA = 20$ and $AB + BC = 21$, then the length of the side BC, equals

(A) 13

(B) 15

(C) 17

(D) 19

$$A = 120^\circ$$

$$a + b = 20$$

$$c + a = 21$$

$$\rightarrow b = 20 - a$$

$$c = 21 - a$$

$$\cos(120^\circ) = \frac{b^2 + c^2 - a^2}{2bc}$$

$$\rightarrow -\frac{1}{2} = \frac{(20-a)(21-a)}{2}$$

$$= (20-a)^2 + (21-a)^2 - a^2$$

$$a^2 - 41a + 420 + a^2 - 42a - 42a + 400 + 441 = 0$$

$$2a^2 - 123a + 1261 = 0$$

$$a = \frac{123 \pm \sqrt{(123)^2 - 4 \cdot 2 \cdot 1261}}{4}$$

$$= \frac{123 \pm \sqrt{15129 - 10088}}{4}$$

$$= \frac{123 \pm \sqrt{5041}}{4} = \frac{123 \pm 71}{4}$$

$$= \frac{194}{4}, 13$$

Ques

In a ΔABC , $BC = 2$, $CA = 1 + \sqrt{3}$ and $\angle C = 60^\circ$. Feet of the perpendicular from A, B and C on the opposite sides BC, CA and AB are D, E and F respectively and are concurrent at P. Now match the entries of **column-I** with respective entries of **column-II**.

Column-I

Column-II

(A) Radius of the circle circumscribing the ΔDEF , is

(P)

$$\frac{\sqrt{6} - \sqrt{2}}{4}$$

$$= \frac{\sqrt{3} - 1}{2\sqrt{2}}$$

(B) Area of the ΔDEF , is

(Q)

$$\frac{1}{\sqrt{2}}$$

(C) Radius of the circle inscribed in the ΔDEF , is

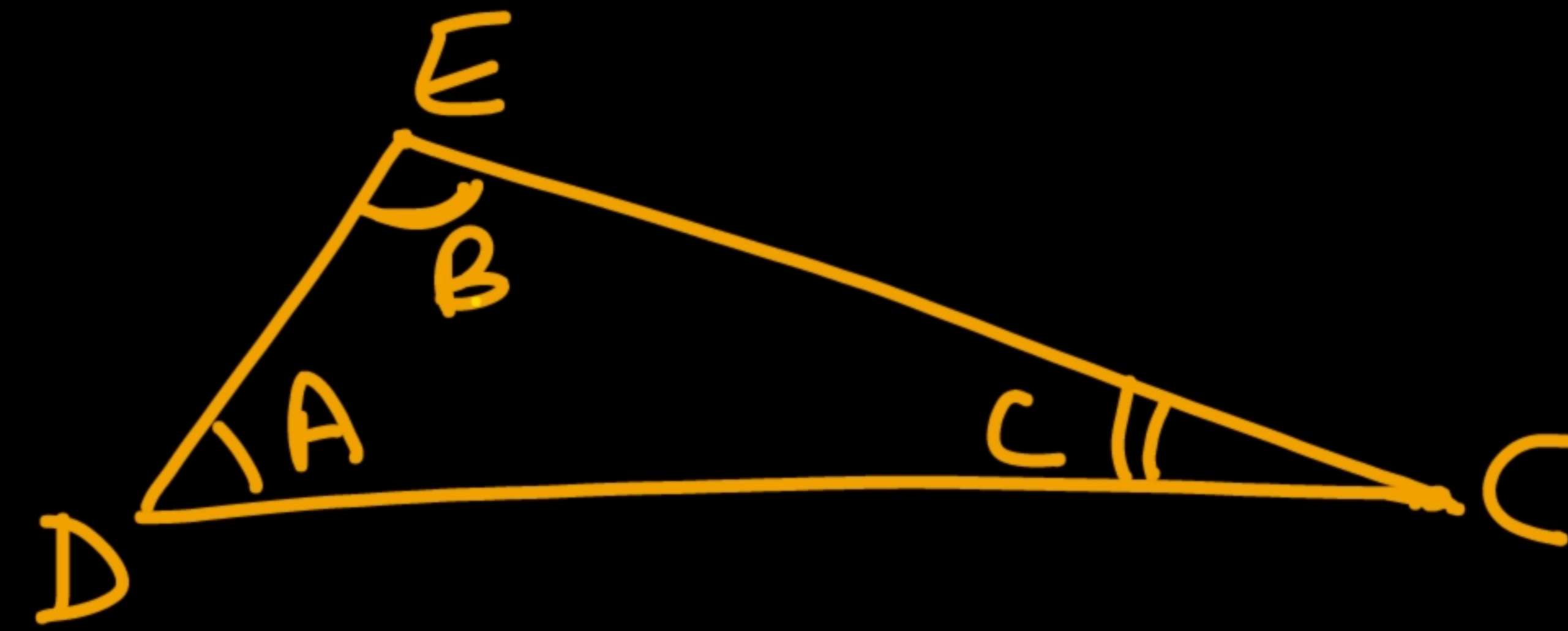
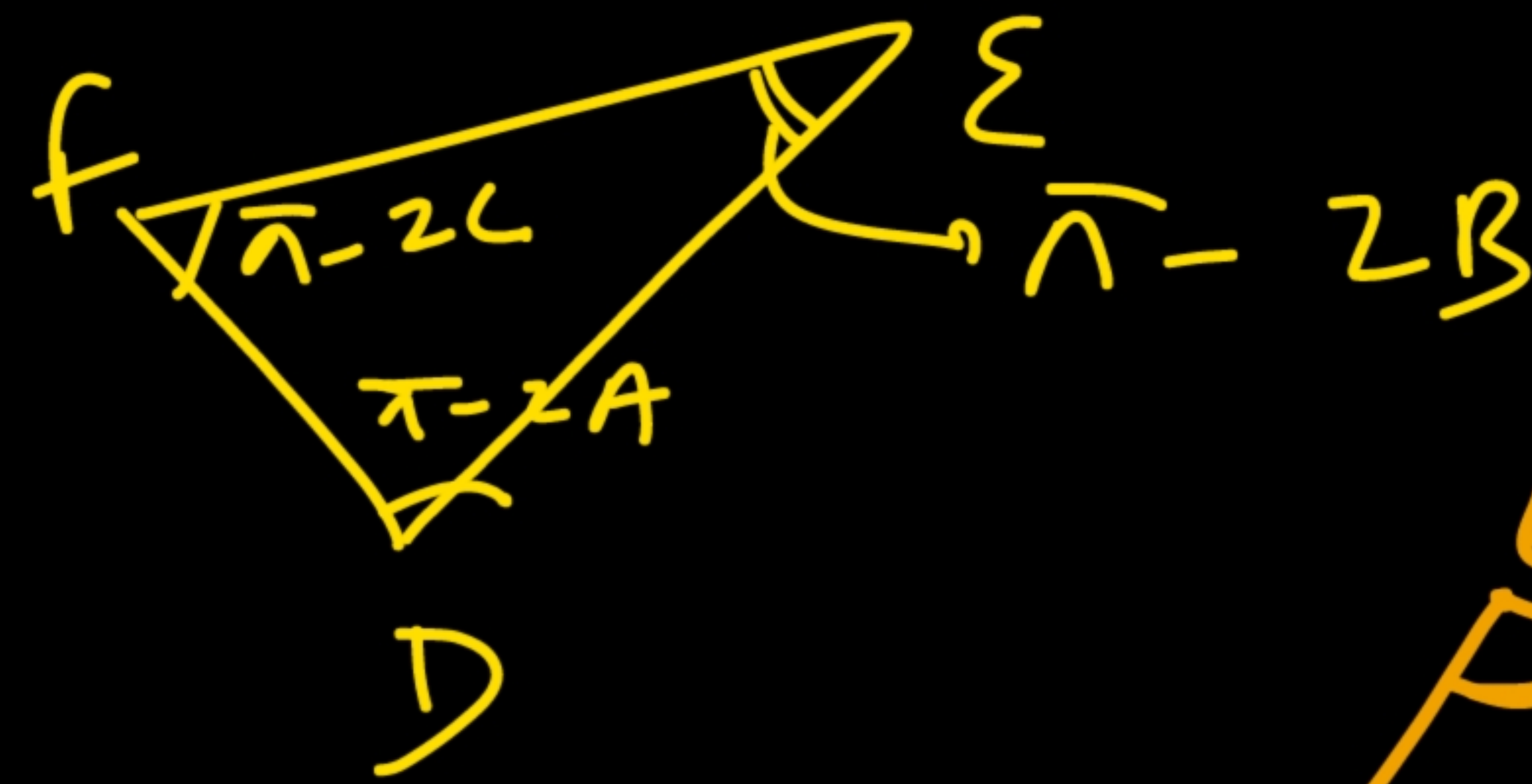
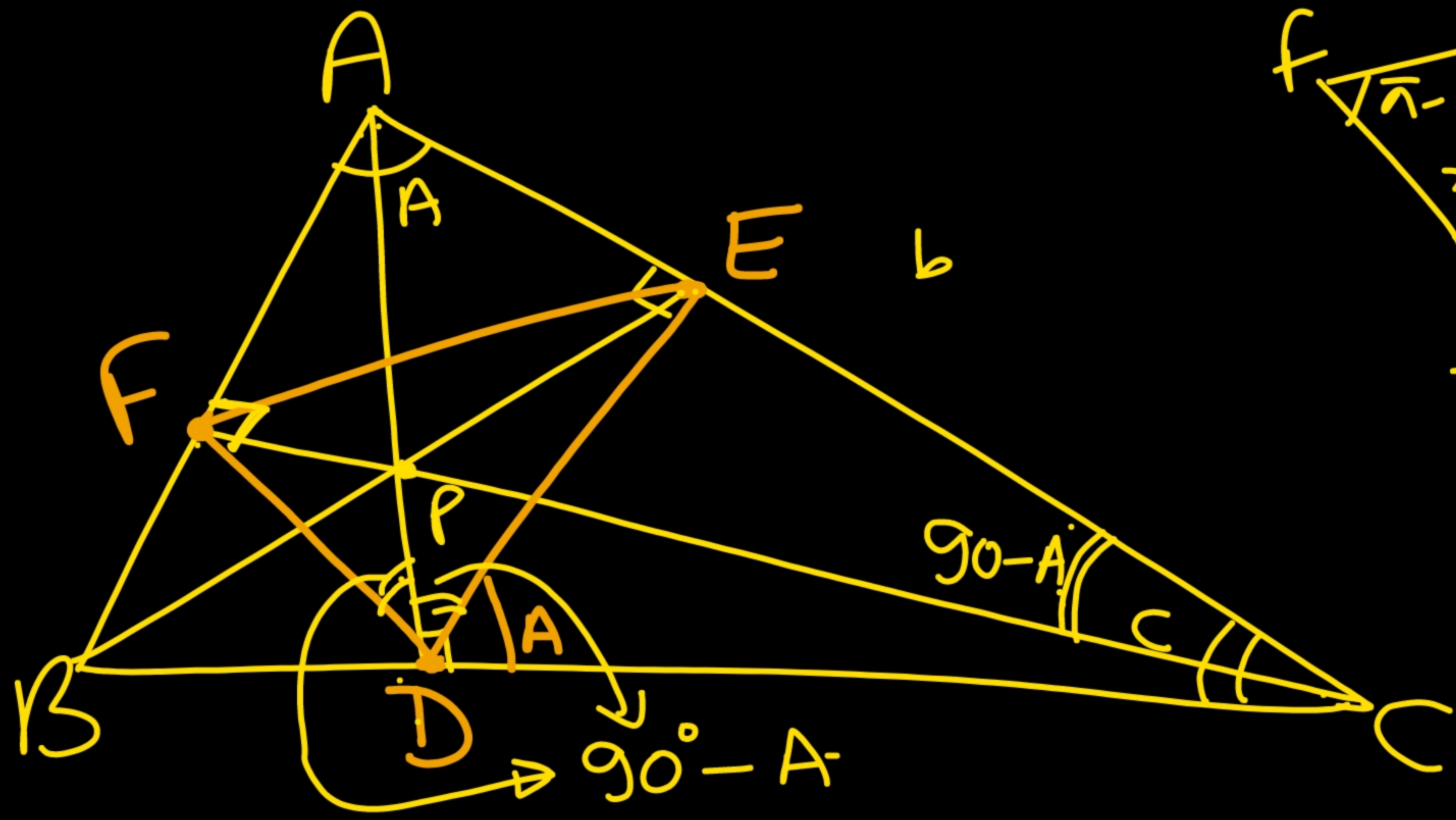
(R)

$$\frac{\sqrt{3}}{4}$$

(S)

$$\frac{\sqrt{6} + \sqrt{2}}{4}$$

$$\begin{aligned} a &= 2 \\ b &= 1 + \sqrt{3} \\ C &= 60^\circ \end{aligned} \Rightarrow c^2 = a^2 - ab + b^2$$
$$= 4 - 2 - 2\sqrt{3} + 4 - 2\sqrt{3}$$
$$= 6$$
$$\boxed{c = \sqrt{6}}$$

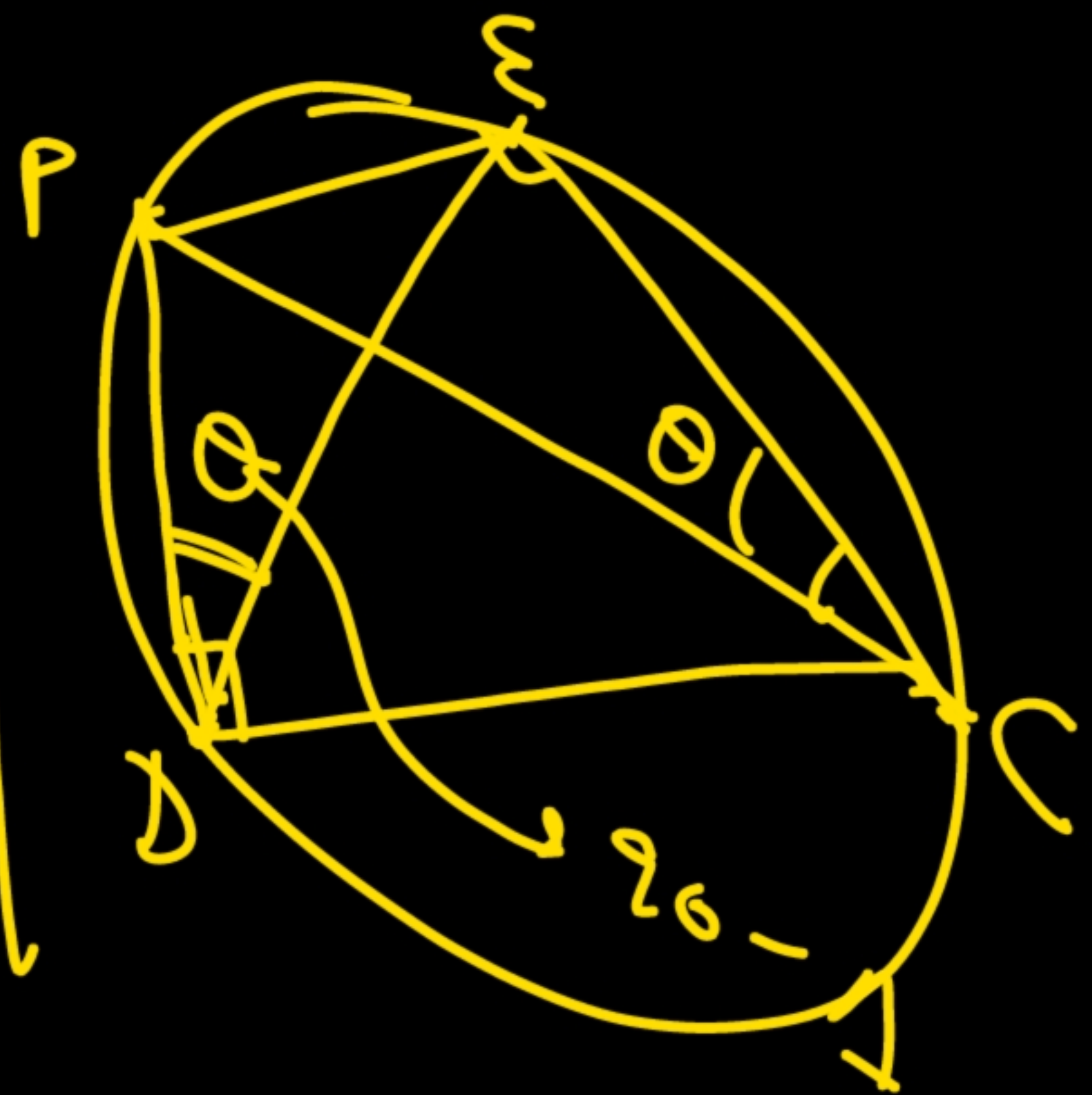


$$\frac{\sin(B)}{DC} = \frac{\sin(C)}{ED}$$

$$ED = \frac{b \cos(C) \cdot \sin(A)}{\sin(B)} = 2R \sin(A) \cos(C)$$

$$\begin{aligned} a &= 2 \\ b &= 1 + \sqrt{3} \\ c &= \sqrt{6} \\ C &= 60^\circ \end{aligned}$$

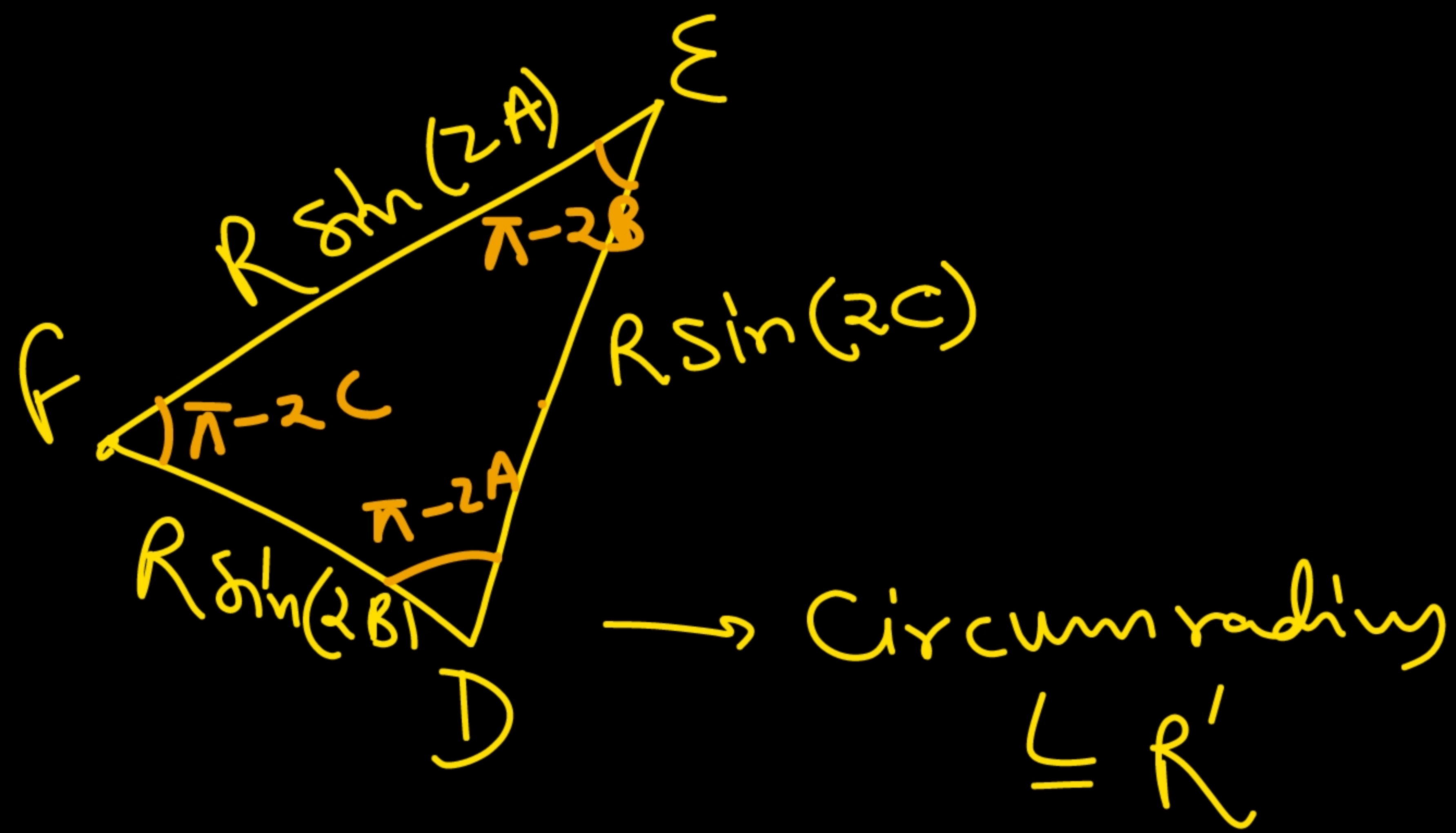
$DC = b \cos(C)$
 $P, D, C, E \rightarrow$ Concyclic



$$\begin{aligned} 2R \sin(C) &= c \\ 2R \frac{\sqrt{3}}{2} &= \sqrt{6} \\ R &= \sqrt{2} \end{aligned}$$

$$ED = R \sin(2C)$$

$$\begin{aligned} \frac{\sin(A)}{a} &= \frac{\sin(C)}{c} \\ \frac{\sin(A)}{2} &= \frac{\frac{1}{\sqrt{3}}}{\sqrt{6}} = \frac{1}{\sqrt{2}} \\ \sin(A) &= \frac{1}{\sqrt{2}} \Rightarrow A = 45^\circ, 135^\circ \end{aligned}$$



$$a = 2R \sin(A) \Rightarrow ED = 2R' \sin(\pi - 2C)$$

$$R \sin(2C) = 2R' \sin(2C)$$

$$R' = \frac{R}{2} = \frac{\sqrt{2}}{2} = \frac{1}{\sqrt{2}}$$

$$\Delta = 2R^2 \sin(A) \sin(B) \sin(C)$$

$$= \frac{abc}{4R}$$

$$\Delta_{DEF} = 2R'^2 \sin(2A) \sin(2B) \sin(2C)$$

$$A = 45^\circ, C = 60^\circ, B = 75^\circ$$

$$\Delta_{DEF} = 2 \cdot \frac{1}{2} \cdot 1 \cdot \frac{1}{2} \cdot \frac{\sqrt{3}}{2} = \frac{\sqrt{3}}{4}$$

$$R = 4R \sin\left(\frac{A}{2}\right) \cdot \sin\left(\frac{B}{2}\right) \sin\left(\frac{C}{2}\right)$$

$$Y_{DEF} = 4R' \sin\left(\frac{\pi - 2A}{2}\right) \cdot \sin\left(\frac{\pi - 2B}{2}\right) \cdot \sin\left(\frac{\pi - 2C}{2}\right)$$

$$= 4 \cdot \frac{1}{\sqrt{2}} \prod \cos(A)$$

$$= \cancel{4} \cdot \frac{1}{\sqrt{2}} \cdot \frac{1}{2} \cdot \frac{1}{\sqrt{2}} \cdot \left(\frac{\sqrt{3}-1}{2\sqrt{2}}\right)$$

$$= \frac{\sqrt{3}-1}{2\sqrt{2}}$$

$$\begin{aligned} & \cos(75^\circ) \\ & \quad = 45^\circ + 30^\circ \\ & = \frac{1}{\sqrt{2}} \cdot \frac{\sqrt{3}}{2} - \frac{1}{\sqrt{2}} \cdot \frac{1}{2} \end{aligned}$$

Que. There is a point (p, q) on the graph of $f(x) = x^2$ and a point (r, s) on the graph of $g(x) = -\frac{8}{x}$ where $p > 0$ and $r > 0$. If the line through (p, q) and (r, s) is also tangent to both the curves, then find the value of $(p+r)$.

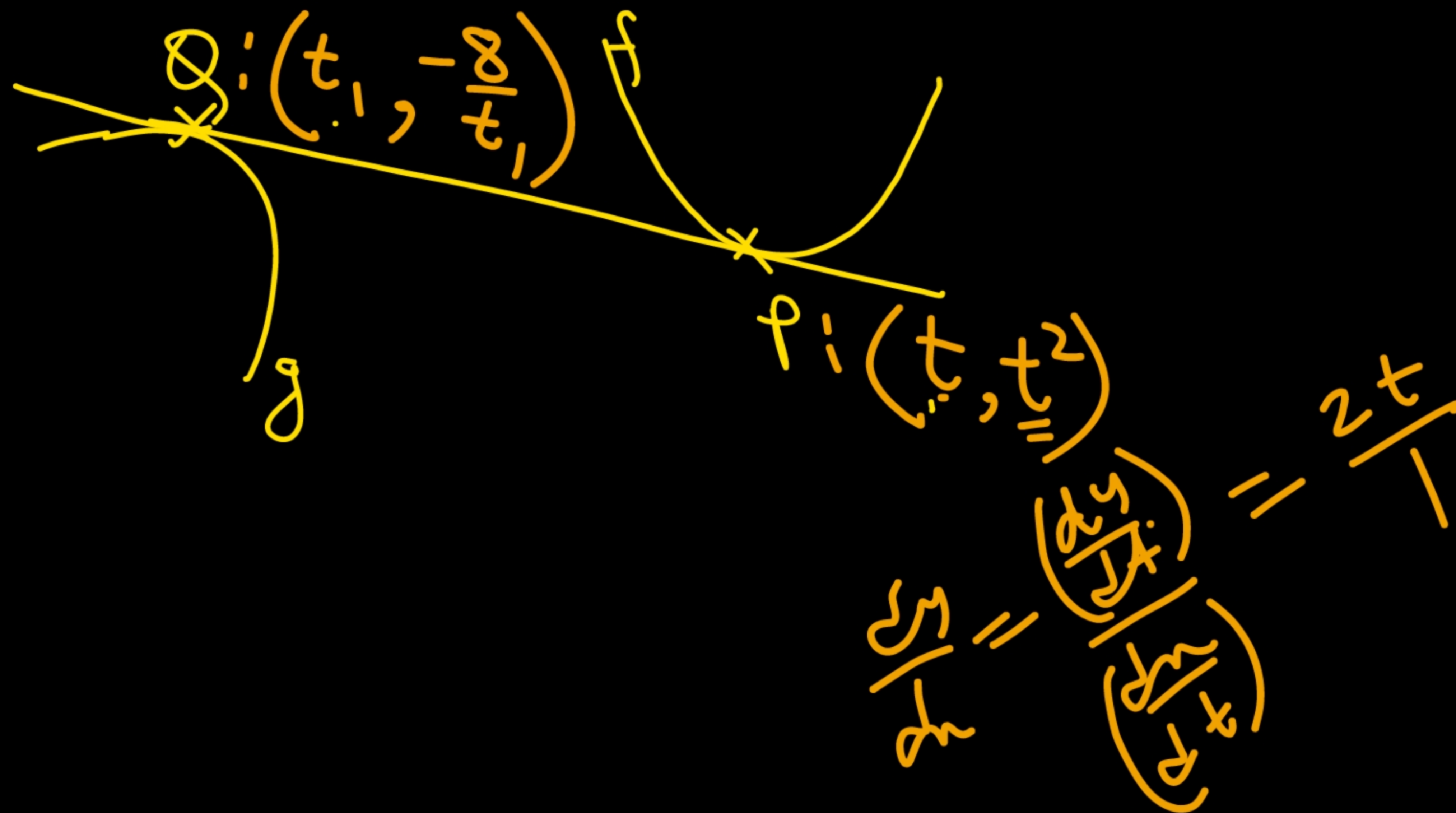
Ans 5

$$f(x) = x^2 \rightarrow P: (p, q)$$

$$g(x) = -\frac{8}{x} \rightarrow Q: (r, s)$$

$$PQ \equiv y - q = 2p(x - p) \leftarrow T_p$$

$$PQ \equiv y + \frac{8}{r} = \frac{8}{r^2}(x - r) \leftarrow T_q$$



$$\begin{aligned} 2t &= \frac{8}{t_1^2} \\ -t^2 &= -\frac{16}{t_1} \Rightarrow t^4 = \frac{16^2}{t_1^2} \end{aligned}$$

$$\div \rightarrow \frac{t^4}{t} = \frac{256}{84} \Rightarrow t^3 = 64 \Rightarrow t = 4$$

$$\Rightarrow t_1 = 1$$

$$p + r = t + t_1 = \boxed{5}$$

Ans

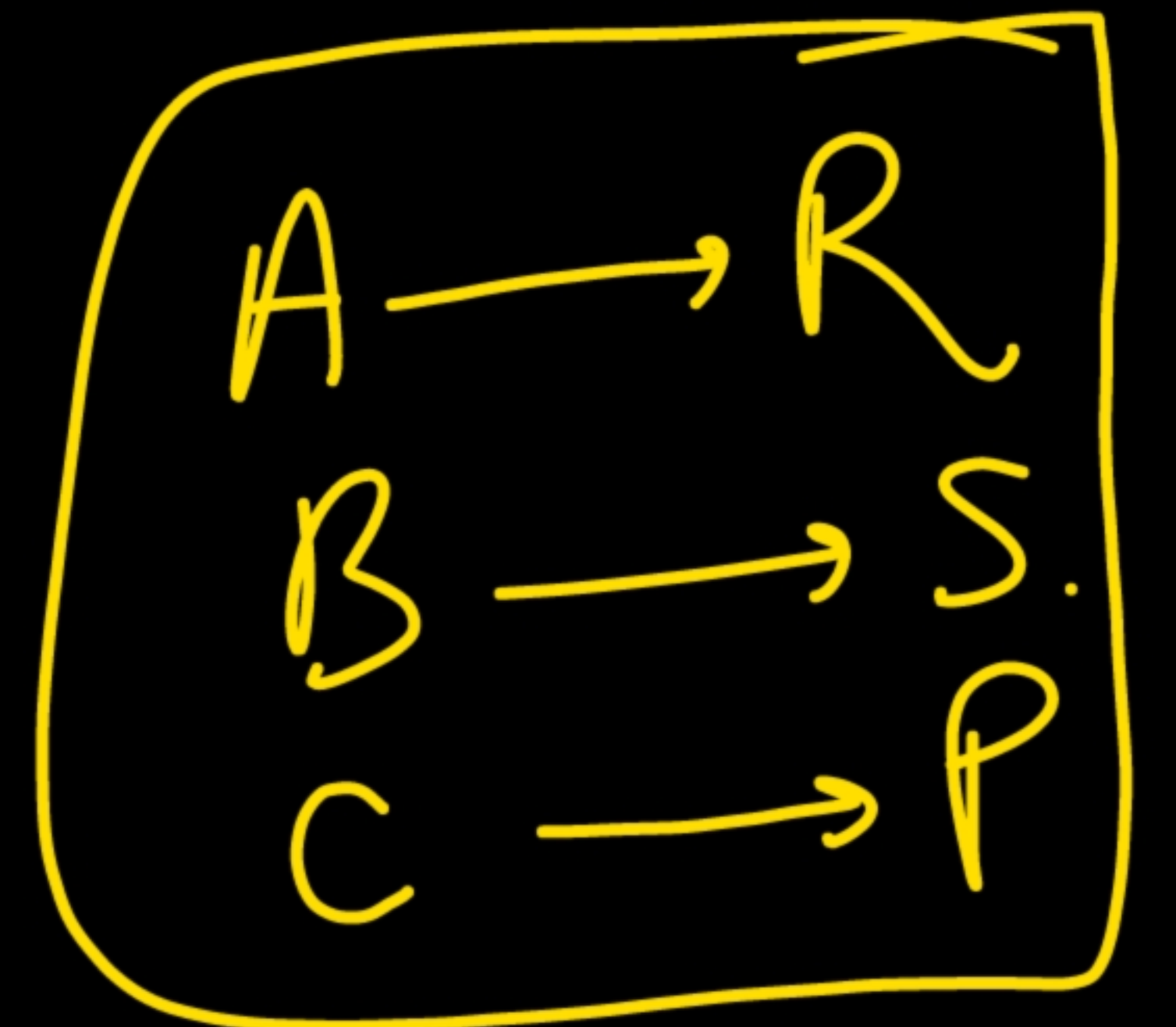
Que

Column-I

Column-II

- (A) The 5th and 8th terms of a geometric sequence of real numbers are 7! and 8! respectively. If the sum to first n terms of the G.P. is 2205, then n equals (R)
- (B) If $x \in (0, 2\pi)$ and $y \in (0, 2\pi)$, then the number of distinct ordered pairs (x, y) satisfying the equation $9\cos^2 x + \sec^2 y - 6\cos x - 4\sec y + 5 = 0$, is (S)
- (C) Let $f(x) = ax^2 + bx + c$ (where $a \neq 0$ and $a, b, c \in \mathbb{R}$) and $(a+c)^2 < b^2$, then the number of distinct values of x in $(-1, 1)$ satisfying the equation $f(x) = 0$ will be (P)

- (P) 1
(Q) 2
(R) 3
(S) 4



$$\begin{aligned} T_5 &= 7! = a \cdot r^4 \\ T_8 &= 8! = a \cdot r^7 \end{aligned} \quad \div \quad r^3 = \frac{8!}{7!} = 8 \Rightarrow r = 2$$

$$\frac{720 \times 7}{45} = a \cdot 16$$

$$a = 315$$

$$a \frac{(r^n - 1)}{r - 1} = 2205$$

$$315 \frac{(2^n - 1)}{2 - 1} = 2205$$

$$2^n = 1 + \frac{2205}{315} = 8$$

$$\boxed{n = 3}$$

$$\begin{aligned} (a+c)^2 &< b^2 \\ (a+c)^2 - b^2 &< 0 \\ (a+b+c)(a-b+c) &< 0 \end{aligned}$$

$$f(1) \cdot f(-1) < 0$$



$$x, y \in (0, 2\pi)$$

$$\underline{9\cos^2(x) + \sec^2(y) - 6\cos(x) - 4\sec(y) + 5 = 0}$$

$$(3\cos(x) - 1)^2 + (\sec(y) - 2)^2 = 0.$$

$$3\cos(x) - 1 = 0 \quad \textcircled{8} \quad \sec(y) = 2.$$

$$\cos(x) = \frac{1}{3} \quad \textcircled{9} \quad \cos(y) = \frac{1}{2}.$$

Ques

If the expression $(1 + ir)^3$ is of the form of $s(1 + i)$ for some real 's' where 'r' is also real and $i = \sqrt{-1}$ then the value of 'r' can be

~~(A)~~ $\cot \frac{\pi}{8}$
 $\underline{= \sqrt{2} + 1}$

~~(B)~~ $\sec \pi$
 $\underline{= -1}$

~~(C)~~ $\tan \frac{\pi}{12}$

~~(D)~~ $\tan \frac{5\pi}{12}$
 $\underline{= \cot(\frac{\pi}{12})}$
 $\underline{= \frac{1}{2 - \sqrt{3}}}$
 $\underline{= 2 + \sqrt{3}}$

$\tan(\frac{\pi}{8}) = \tan(22\frac{1}{2}^\circ)$
 $= \sqrt{2} - 1$
 $\tan(\frac{\pi}{12}) = \tan(15^\circ)$
 $= 2 - \sqrt{3}$

BCD

$(1 + ir)^3 = s(1 + i)$, $s, r \in \mathbb{R}$
 $1 - ir^3 + 3ir(1 + ir) = s + si$

$1 - 3r^2 + i(3r - r^3) = s + is$

$1 - 3r^2 = s$

$3r - r^3 = s$

$1 - 3r^2 = 3r - r^3$

$r^3 - 3r^2 - 3r + 1 = 0$

$\Rightarrow (r + 1)(r^2 - 4r + 1) = 0$

$\Rightarrow r = -1, \frac{4 \pm \sqrt{12}}{2}$
 $= -1, 2 \pm \sqrt{3}$

Qwe

For $x \geq 0$, the smallest value of the function $f(x) = \frac{4x^2 + 8x + 13}{6(1+x)}$, is

(A) 1

☒ (B) 2

(C) $\frac{25}{12}$

(D) $\frac{13}{6}$

$x \geq 0$

$$f(x) = \frac{4x^2 + 8x + 13}{6(1+x)}$$

$$f(x) = \frac{4(x+1)^2 + 9}{6(1+x)}$$

$1+x = t \geq 1$

$$g(t) = \frac{4t^2 + 9}{6t}$$

$$= \frac{1}{6} \left(4t + \frac{9}{t} \right) \geq \frac{1}{6} \cdot 2\sqrt{4 \cdot 9} = \boxed{2} \checkmark$$

for equality

$$4t = \frac{9}{t} \Rightarrow t^2 = \frac{9}{4} \Rightarrow t = \pm \frac{3}{2}$$

$\Downarrow$
 $g(t)_{\min} = 2$
 $= (f(x))_{\min}$

Que

Let f be a continuous function on $[0, 1]$, differentiable in $(0, 1)$ such that $f(1) = 0$. Then there exists some $c \in (0, 1)$ such that

~~(A) $c f'(c) + f(c) = 0$~~

~~(C) $f'(c) + c f(c) = 0$~~

~~(B) $c f'(c) - f(c) = 0$~~

~~(D) $f'(c) - c f(c) = 0$~~

Continuous in $\rightarrow [0, 1]$

diff in $\rightarrow (0, 1)$

$f(1) = 0$

(A) $x f'(x) + f(x) = 0 \rightarrow g(x) = x \cdot f(x)$

(B) $x f'(x) - f(x) = 0$

$\int \downarrow$
 $g(x) = \frac{f(x)}{x}$

$\hookrightarrow c$
 $\hookrightarrow \text{Diff}$

$g(0) = g(1)$

Using R.T.

$\exists c \in (0, 1)$ s.t.

$g'(c) = 0$

$c \cdot f'(c) + f(c) = 0$

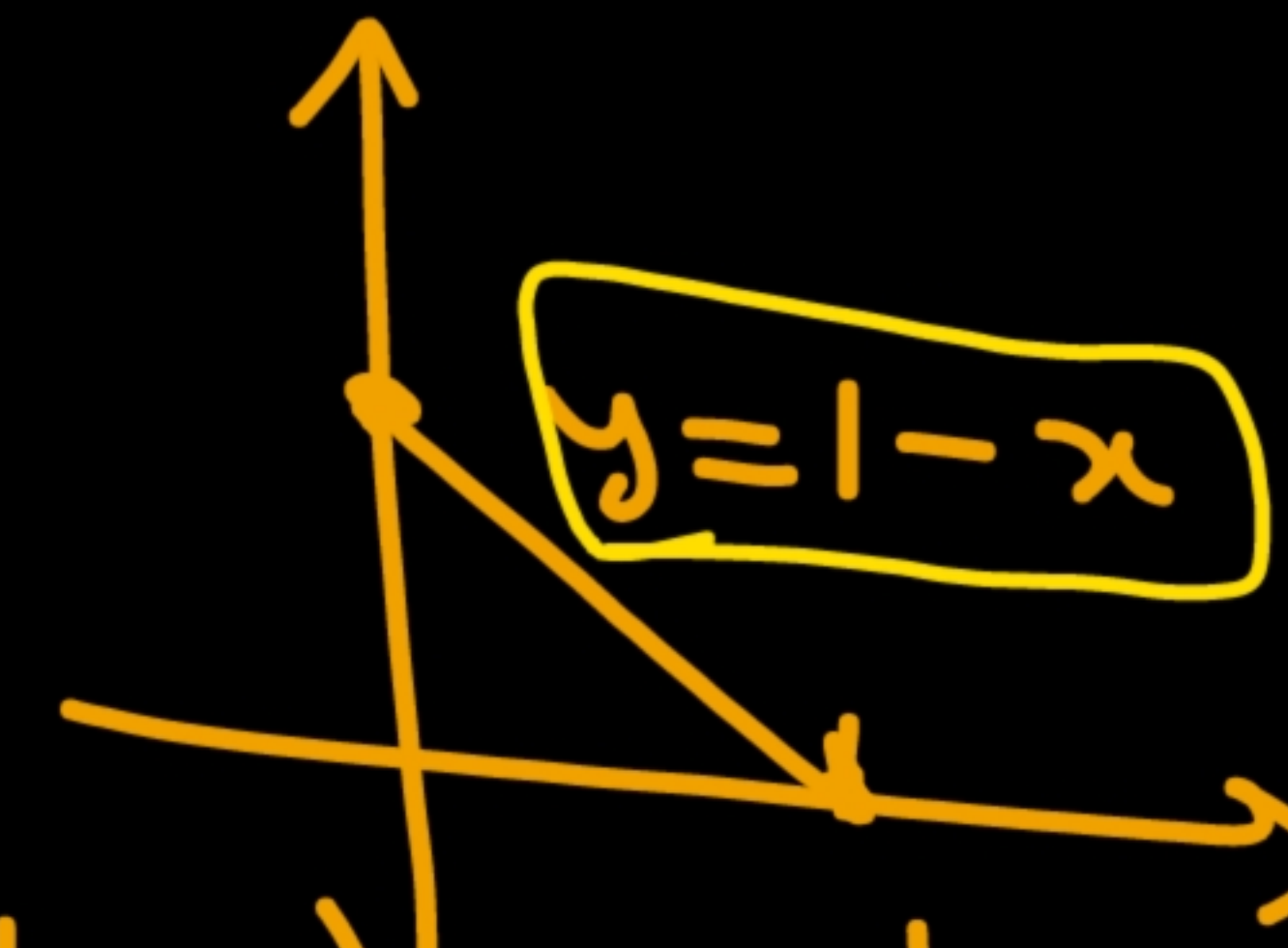
(B)

(C)

$-1 + c(1-c) = 0$
 $c^2 - c + 1 = 0$

~~$f(x) = 0$~~

$c \cdot (-1) - (1-c) = 0$
 $-c - 1 + c = 0$
 $-1 = 0$



$-1 - c(1-c) = 0$
 $-1 - c + c^2 = 0$
 $c^2 - c - 1 = 0$
 $c = \frac{1 \pm \sqrt{5}}{2}$